

Intel 80386 Microprocessor

1 Introduction

The **Intel 80386** is a 32-bit microprocessor and a major advancement in the x86 family. It introduces 32-bit internal and external data buses, supports large memory addressing, and operates in multiple modes while maintaining upward compatibility with earlier x86 processors.

2 Key Features of Intel 80386

- 32-bit microprocessor with 32-bit internal and external data bus
- 32-bit registers
- Supports 8-bit, 16-bit, and 32-bit operands
- 32-bit address bus
- Can address up to **4 GB of physical memory**

3 Memory Organization

- Physical memory is organized into segments
- Maximum segment size is **4 GB**
- Supports up to **16K segments**
- Total virtual memory space:

$$4 \text{ GB} \times 16\text{K} = 64 \text{ TB}$$

4 Prefetch Queue

- Prefetch queue size is **16 bytes**
- Improves instruction fetch and execution efficiency

5 Clock Speed

- Supported clock speeds range from:
 - 16 MHz
 - 20 MHz
 - Up to 33 MHz

6 Operating Modes

The Intel 80386 operates in the following modes:

- **Real Mode**
- **Protected Mode**
- **Virtual Real Mode**

7 Memory Management Unit (MMU)

The Intel 80386 includes an on-chip Memory Management Unit (MMU) responsible for handling memory addresses.

7.1 Segmentation Unit

- Provides protection for data and code in memory
- Supports **four privilege levels** (protection rings):
 - Highest: Operating system kernel
 - High: Device drivers
 - Medium: System utilities
 - Lowest: User applications

7.2 Paging Unit

- Operates only in protected mode
- Converts linear addresses into physical addresses
- Provides virtual memory support
- Enables memory isolation

8 Instruction Set and Compatibility

- Instruction set is upward compatible with earlier x86 processors
- Supports the Intel 80387 numeric data processor

9 Advantages of Intel 80386

- True 32-bit processing capability
- Large physical and virtual memory support
- Advanced memory management with paging and segmentation
- Multiple operating modes
- Improved performance compared to earlier processors

10 Applications

- Advanced multitasking operating systems
- Systems requiring large memory support
- High-performance computing systems
- Protected and virtual memory-based systems

11 Major Differences Between Intel 80286 and Intel 80386

Feature	Intel 80286	Intel 80386
Processor Architecture	16-bit microprocessor	32-bit microprocessor
Data Bus and Registers	16-bit data bus and registers	32-bit data bus and registers
Address Bus	24-bit	32-bit
Maximum Physical Memory	16 MB	4 GB
Operating Modes	Real mode and protected mode	Real mode, protected mode, and virtual real mode
Memory Management	Basic memory management	Advanced memory management with segmentation and paging
Paging Support	Not supported	Supported
Virtual Memory	Not supported	Supported
Clock Speed	Up to 12.5 MHz	16 MHz to 33 MHz